

Group No. A2

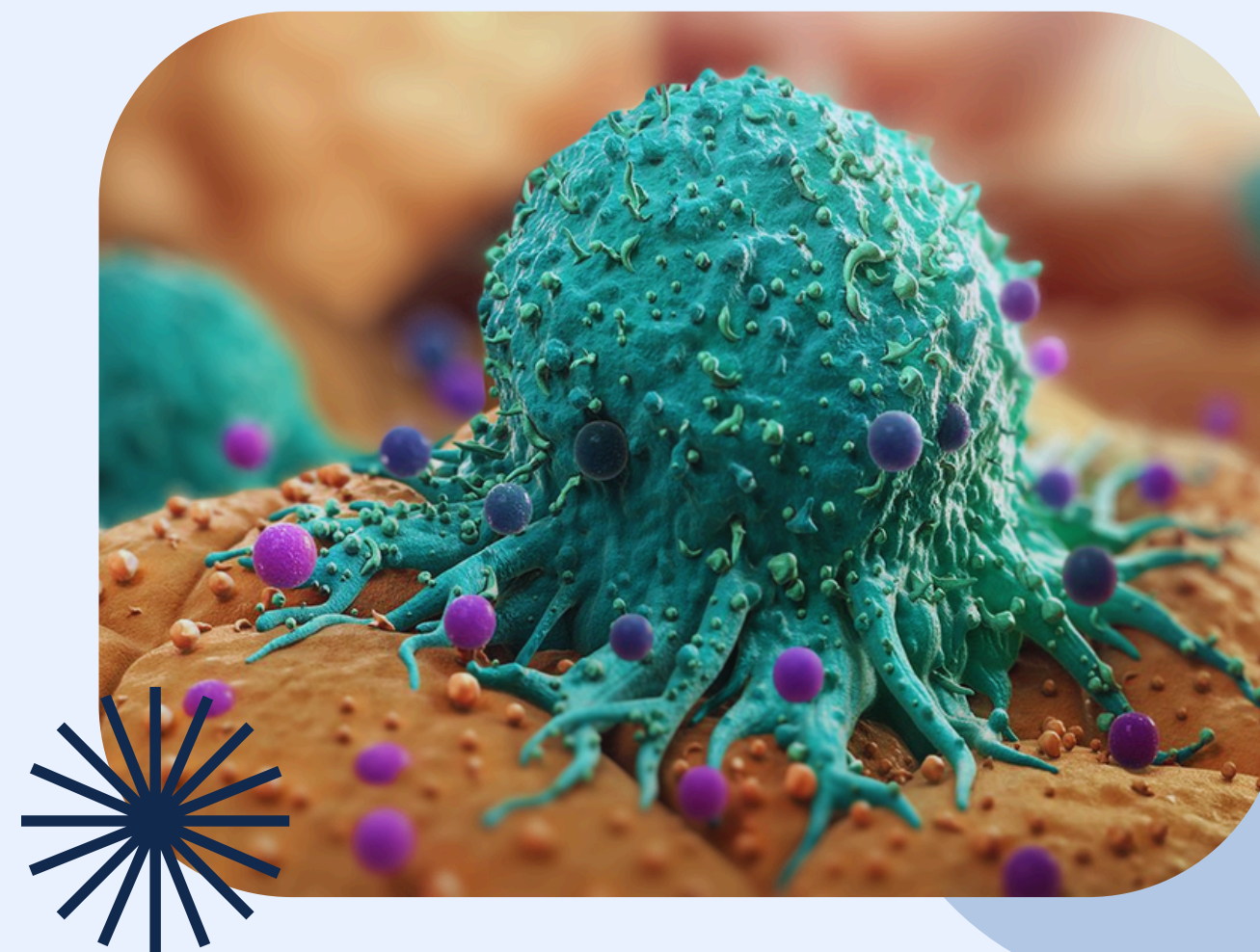
Breast Cancer Classification from Histopathology

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Index

01 Introduction

02 What is Histopathology

03 Who are Histopathologist

04 Nanavati Max Hospital

05 Gratitude to Doctor

06 Breast Cancer

07 Problem Statement

08 Dataset

09 Dataset and Type

10 Benign v/s Malignant

11 Literature Survey

12 Methodology

13 Diagram

14 What is DML

15 What is KL Divergence

16 Results and Discussions

17 Poster

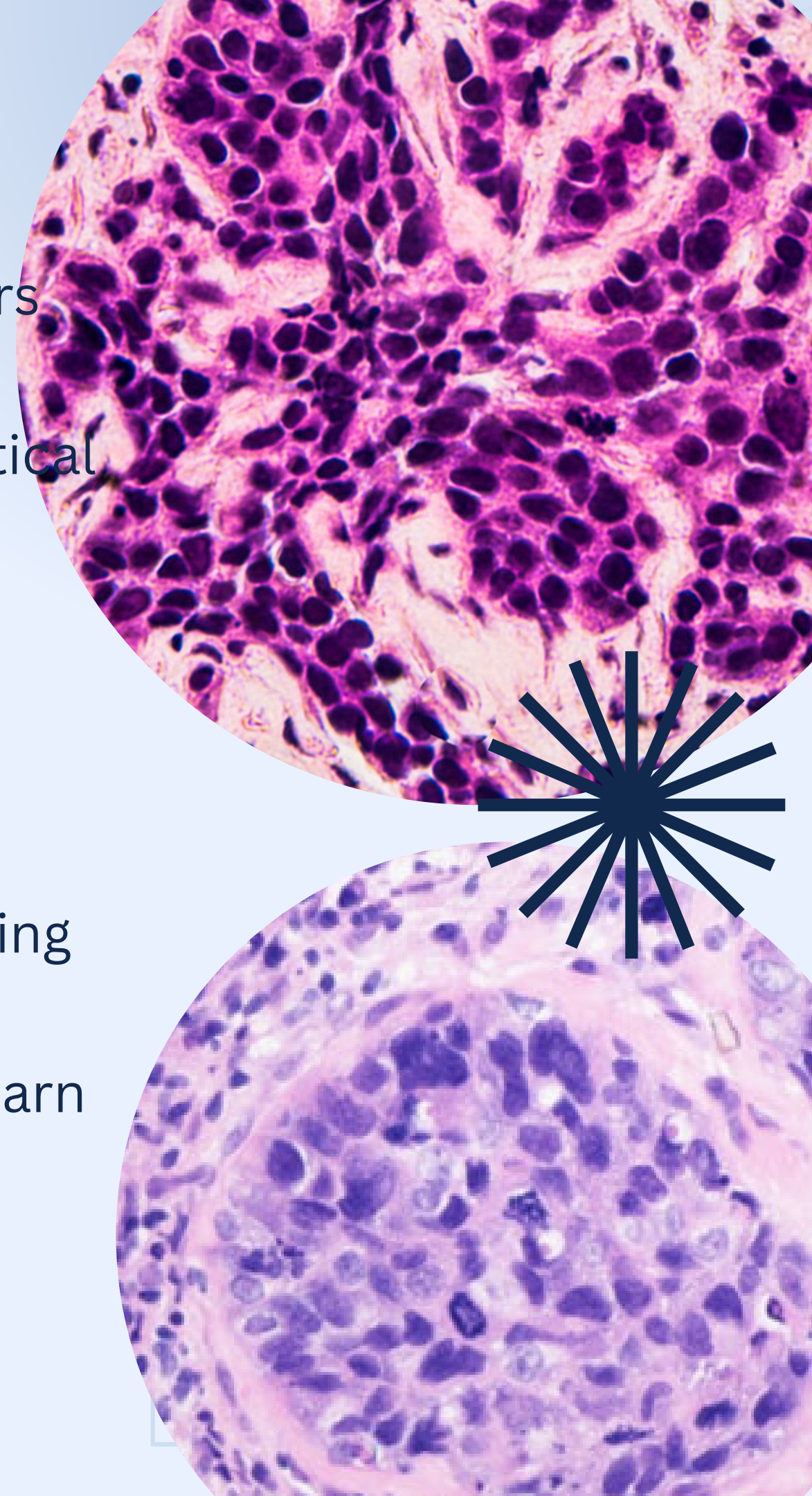
18 Poster Competition

19 Future Scope

20 Conclusion

Introduction

- Breast cancer is one of the most common and life-threatening cancers among women worldwide.
- Early and accurate diagnosis through histopathological images is critical for effective treatment.
- Manual analysis of biopsy slides is time-consuming and prone to variability among pathologists.
- Artificial Intelligence (AI) and Deep Learning have shown significant promise in medical image analysis.
- This project focuses on automated classification of breast cancer using histopathological images.
- We employ Deep Mutual Learning (DML), where two models (CNN) learn collaboratively.
- Kullback–Leibler (KL) Divergence is used to encourage knowledge sharing between models during training

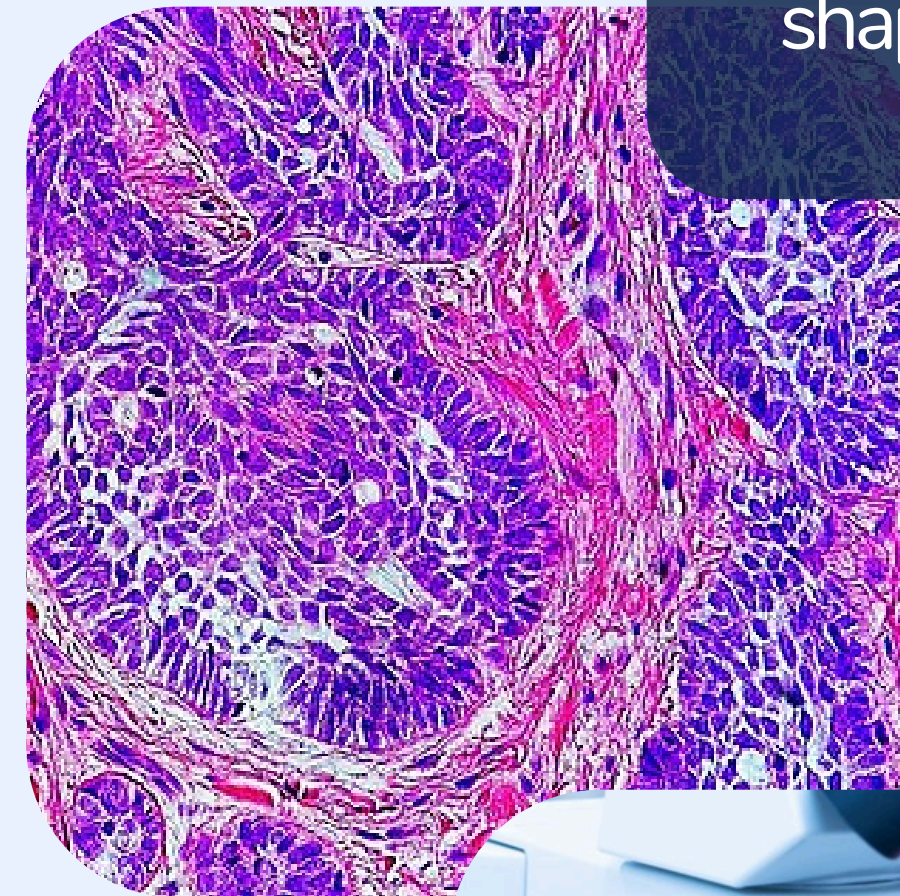


What is Histopathology?

Histopathology is the study of diseased tissue under a microscope.

- It involves examining biopsy samples to identify abnormalities in cells and tissue structure.
- Pathologists use stained tissue slides to detect the presence and type of diseases like cancer.
- In breast cancer diagnosis, histopathology helps determine if the tumor is benign (non-cancerous) or malignant (cancerous).
- It plays a critical role in medical diagnosis, treatment planning, and prognosis.

Trabecular, elongated (rod-shaped) groups of cells



Who are Histopathologist?

A histopathologist is a medical doctor who specializes in diagnosing diseases by examining tissues and cells under a microscope. They play a vital role in detecting conditions like cancer, infections, and inflammatory diseases. By analyzing biopsy samples, they provide critical information that helps doctors plan effective treatments. Histopathologists often collaborate closely with surgeons and oncologists, making them key contributors to patient care.



Nanavati Max Super Speciality Hospital Mumbai

- **Location:** Vile Parle (West), Mumbai, Maharashtra, India
- **Established:** 1950 (originally as Dr. Balabhai Nanavati Hospital)
- **Specialties:** Offers comprehensive services in
 - Oncology (Cancer care)
 - Neurology and Neurosurgery
 - Orthopedics and Joint Replacement
 - Cardiology and Cardiothoracic Surgery
 - Gastroenterology
 - Urology
 - Pediatrics
 - Organ Transplants (e.g., liver, kidney)
- **Advanced Tech:** Equipped with modern diagnostic and treatment technologies including robotic surgery, PET-CT, and MRI
- **Doctors:** Hosts a large team of senior consultants and medical experts across 55+ specialties
- **Research & Training:** Involved in clinical research, medical training, and education
- **Patient Services:** Caters to both domestic and international patients
- **Accreditation:** NABH and NABL accredited for quality and lab services



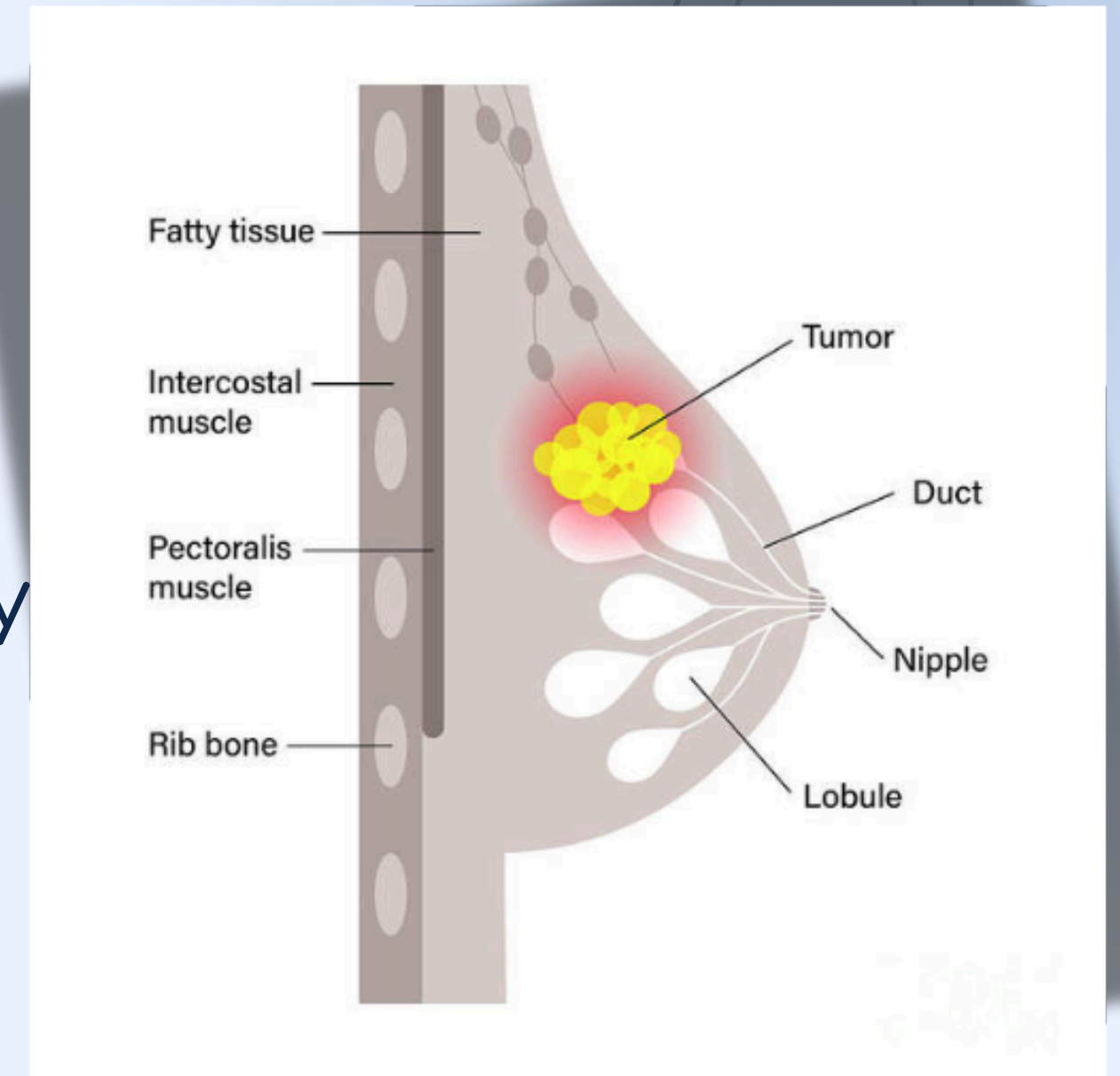
With Sincere Gratitude to **Dr. Geeta Verma**

We would like to thank Dr. Geeta Verma for her invaluable support and guidance throughout our project. The insights into cancerous tissues greatly enriched our understanding and helped us navigate the complexities of histopathology with confidence. Her time and expertise made a meaningful impact, and we are truly grateful for her contribution.



Breast Cancer

- Breast cancer is a disease in which abnormal cells in the breast grow uncontrollably.
- It commonly begins in the ducts (milk-carrying tubes) or lobules (milk-producing glands) of the breast.
- It can be benign (non-cancerous) or malignant (cancerous)—malignant tumors can invade nearby tissues or spread to other parts of the body.
- Early detection through screening and diagnosis improves survival rates.
- Histopathology is a key method used to classify and understand the nature of breast tumors.

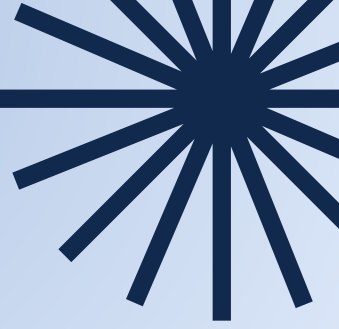


Problem Statement

1. Histopathological analysis currently relies heavily on manual assessment by pathologists.
2. Manual analysis can lead to variability, delays, and human errors in diagnoses.
3. Pathologists are experiencing increasing workloads due to rising demands for tissue-based diagnoses.
4. Time-consuming processes affect the speed and efficiency of diagnostic workflows.



Objectives



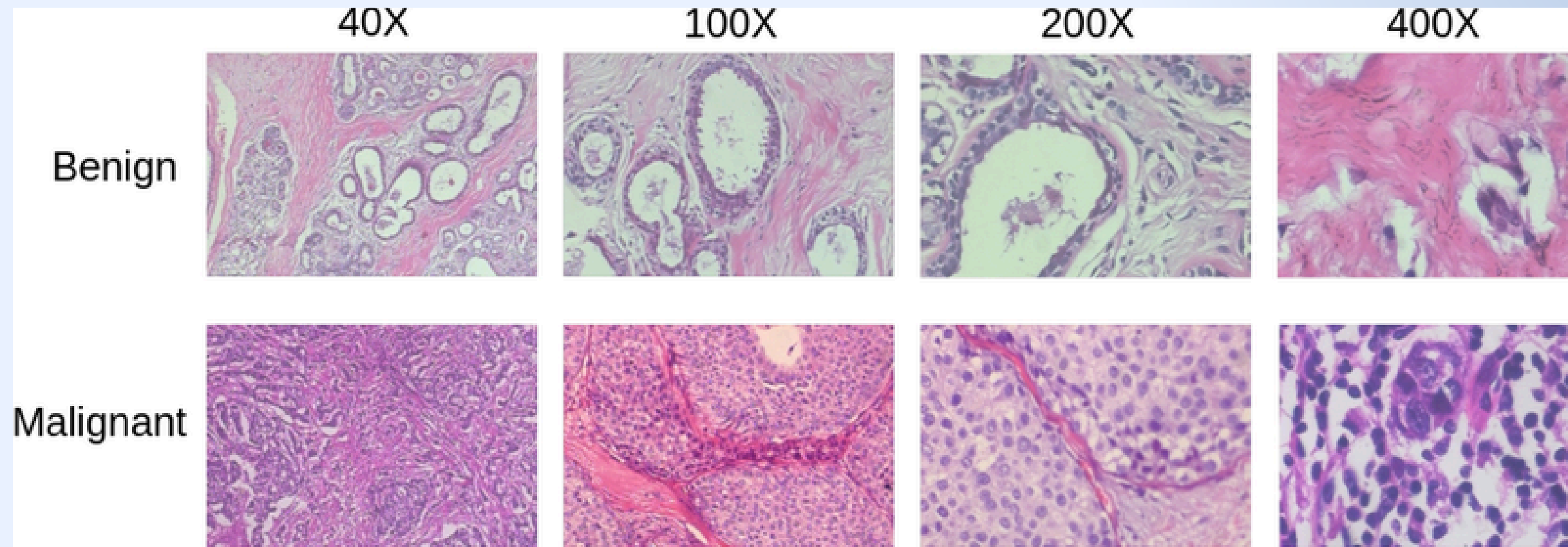
- Achieve Time-Efficient Diagnosis (a) Develop an AI-based system that accelerates the analysis of histopathological images, reducing diagnostic delays. (b) Minimize the time pathologists spend on routine image evaluations, allowing them to focus on complex cases.
- Ensure Diagnostic Accuracy and Consistency (a) Improve diagnostic precision by minimizing human errors associated with manual assessments. (b) Provide consistent results across different datasets and clinical settings.
- Build a Robust and Diverse Dataset for Training (a) Create a comprehensive dataset of histopathological images representing both benign and malignant tissues. (b) Ensure the dataset reflects diverse conditions and demographic variations to improve model performance.
- Design an AI Model for Early and Reliable Detection (a) Develop an AI model that accurately identifies and differentiates between normal and abnormal tissues. (b) Detect subtle patterns and features for early-stage breast cancer diagnosis, improving patient outcomes.
- Create a User-Friendly System for Pathologists (a) Design an intuitive interface that integrates seamlessly into pathologists' work-flows. (b) Ensure the system provides actionable insights while keeping pathologists in control of final diagnostic decisions.
- Validate Model Generalizability Across Diseases (a) Test the AI system on various datasets to ensure its reliability across different diseases and patient populations. (b) Confirm that the model performs well in diverse clinical environments, enhancing its real-world applicability.

Dataset

Malignant

Benign

Dataset and Type



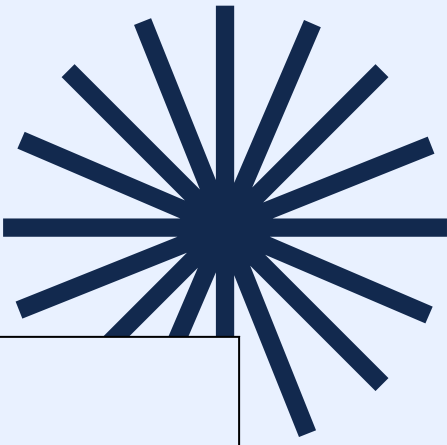
- Magnification shows how much larger a sample appears compared to its actual size.
- Common Magnifications: 40x, 100x, 200x, and 400x

Total Images : 7909
Training : 6406
Validation : 712
Test : 791

Train Malignant :4400
Train Benign :2006

We received histopathological image data with only 40X magnification from the hospital. Therefore, our entire project is based exclusively on the 40X magnification factor.

Benign v/s Malignant



Benign tumors	Malignant tumors
Not cancerous	Cancerous
Don't invade surrounding tissue	May invade surrounding tissue
Don't spread to other parts of the body	May spread to other parts of the body
Grow slowly	Grow quickly
Are not likely to recur	Are more likely to recur
Have a smooth, regular shape	May have an uneven shape
Move around when pushed on	Don't move around when pushed on
May or may not require treatment	Require treatment

WHAT ARE WE PREDICTING

Our AI model predicts the type of tissue in histopathological images: Benign (Non-cancerous) Malignant (Cancerous) Goal: Classify breast tissue samples based on microscopic image features. Helps in: Early detection of breast cancer Supporting pathologists in diagnosis Improving accuracy and speed of cancer classification Output: A binary classification — Benign or Malignant

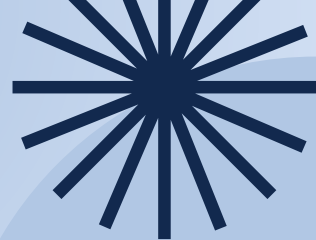


LITERATURE SURVEY



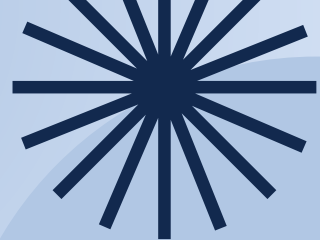
Sr.no	Author	Title	Year	Summary
1	Kaur, A.; Kaushal, C.; Sandhu, J.K.; Dama'sevi'cius, R.; Thakur, N.	Histopatho- logical Image Diagnosis for Breast Cancer Diagnosis Based on Deep Mutual Learning	2024	Proposes Deep Mutual Learning for breast cancer diagnosis; achieves high accuracy and better interpretability.
2	S. Sundararajan and R. Ganesan	Leukocyte Extraction Using Deep CNN	2023	Uses CNNs (VGG-UNet, U-Net, SegNet) for leukocyte segmentation; outperforms traditional methods.

LITERATURE SURVEY



Sr.no	Author	Title	Year	Summary
3	A. Belharbi, A. Ben-Hamza, C. B. Santiago, H. Mouch`ere, and T. G´eraud	Weakly Supervised Object Localization in Histology Images	2022	Reviews WSOL methods; addresses challenges like CAM under-activation in cancer diagnosis.
4	Saad Awadh Alanazi, M. M. Kamruzzaman, Md Nazirul Islam Sarker, Madal- lah Alruwaili, Yousef Alhwaiti	Boosting Breast Cancer Detection Using CNN	2021	Uses CNNs (VGG- UNet, U-Net, SegNet) for leukocyte segmentation; outperforms traditional methods.

LITERATURE SURVEY



Sr.no	Author	Title	Year	Summary
5	Diwakaran, M., Surendran, D	Breast Cancer Prognosis Based on Trans-fer Learning Techniques in Deep Neural Networks	2023	Introduces BCP-TL model; Xception + Channel Boosted CNN achieves 98.96% accuracy.
6	Sharma, S. and Mehra, R.	Mehra Conventional Machine Learning and Deep Learning Approach for Multi Classification of Breast Cancer Histopathology Images	2020	Compares handcrafted and transfer learning methods; VGG16+SVM performs best.



METHODOLOGY :

- **Dataset Collection & Preprocessing**

- Histopathological images of breast tissue (train/test split)

- Image resizing, normalization, and augmentation

- **Model Selection** of two deep learning models: CNN

- **Deep Mutual Learning (DML)**

- Both models are trained simultaneously

- They share knowledge by learning from each other's outputs

- **KL Divergence Loss**

- Used to measure the difference between model predictions

- Encourages both models to align and improve together

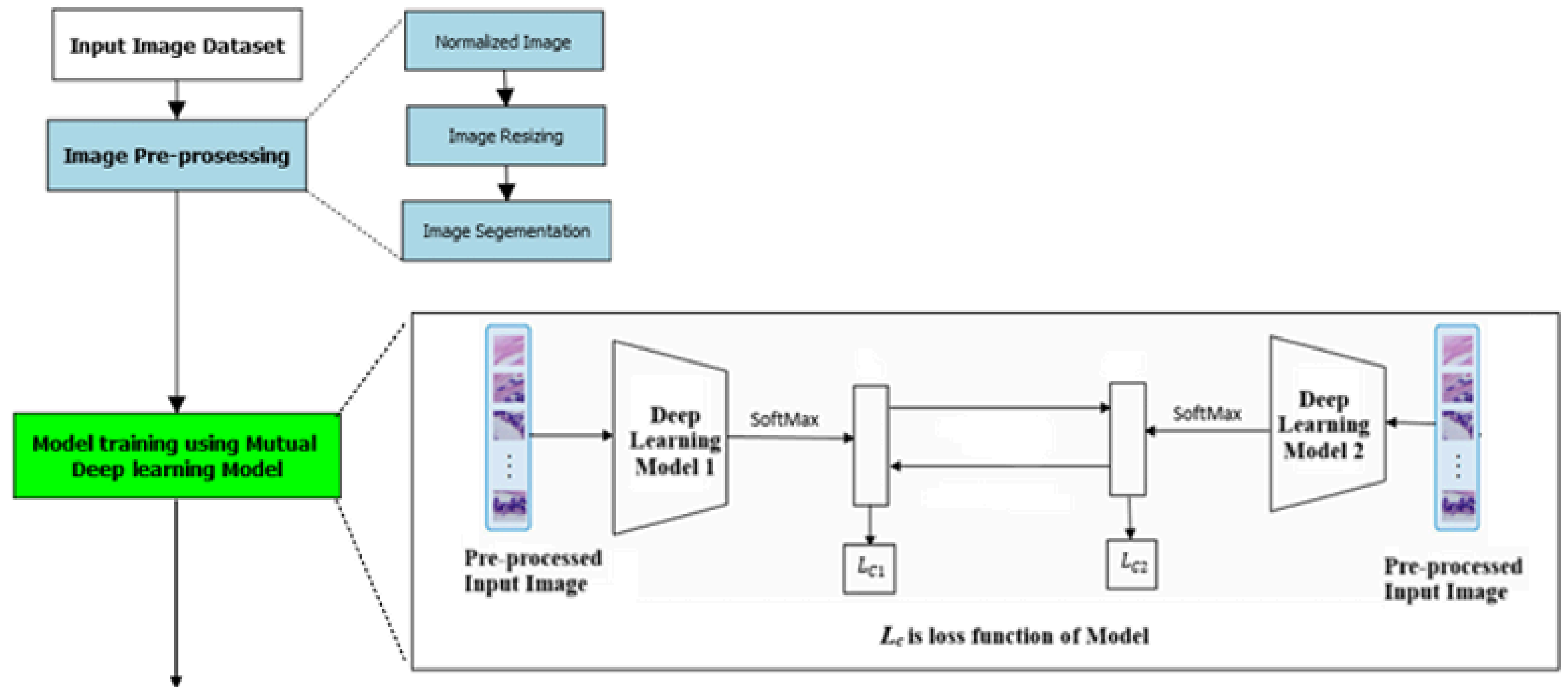
- **Training & Evaluation**

- Models trained on GPU with batch learning

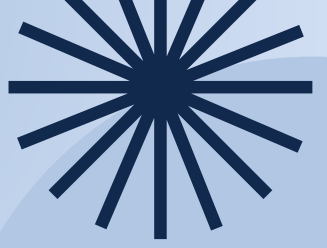
- Evaluated using accuracy, precision, recall, and F1-score

- **Final Output** : Predict whether the input image is Benign or Malignant



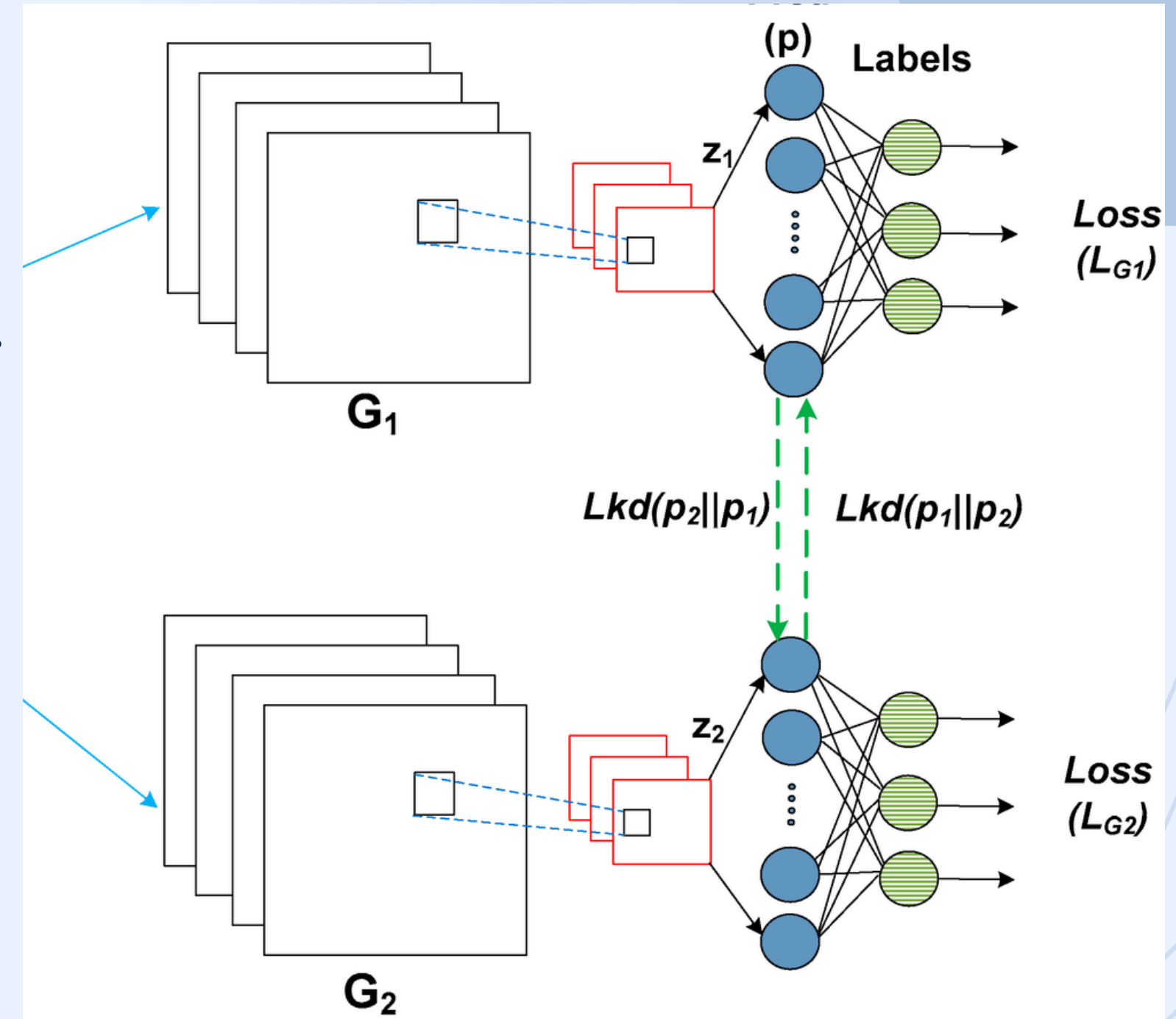


Output
(Normal Image or malignant Image)



WHAT IS DML (DEEP MUTUAL LEARNING)?

- A technique where two or more neural networks learn collaboratively instead of independently.
- Unlike traditional training (teacher-student), each model acts as both teacher and student.
- During training, models exchange knowledge by sharing their predictions.
- Each model not only minimizes its own classification loss but also minimizes KL Divergence between its predictions and the peer model's predictions.
- This leads to better generalization, reduced overfitting, and improved performance.



WHAT IS KL DIVERGENCE ?

- KL Divergence (Kullback–Leibler Divergence)
- A mathematical measure of how one probability distribution is different from another
- In Deep Learning, it is used to measure how close the predicted probabilities of two models are.

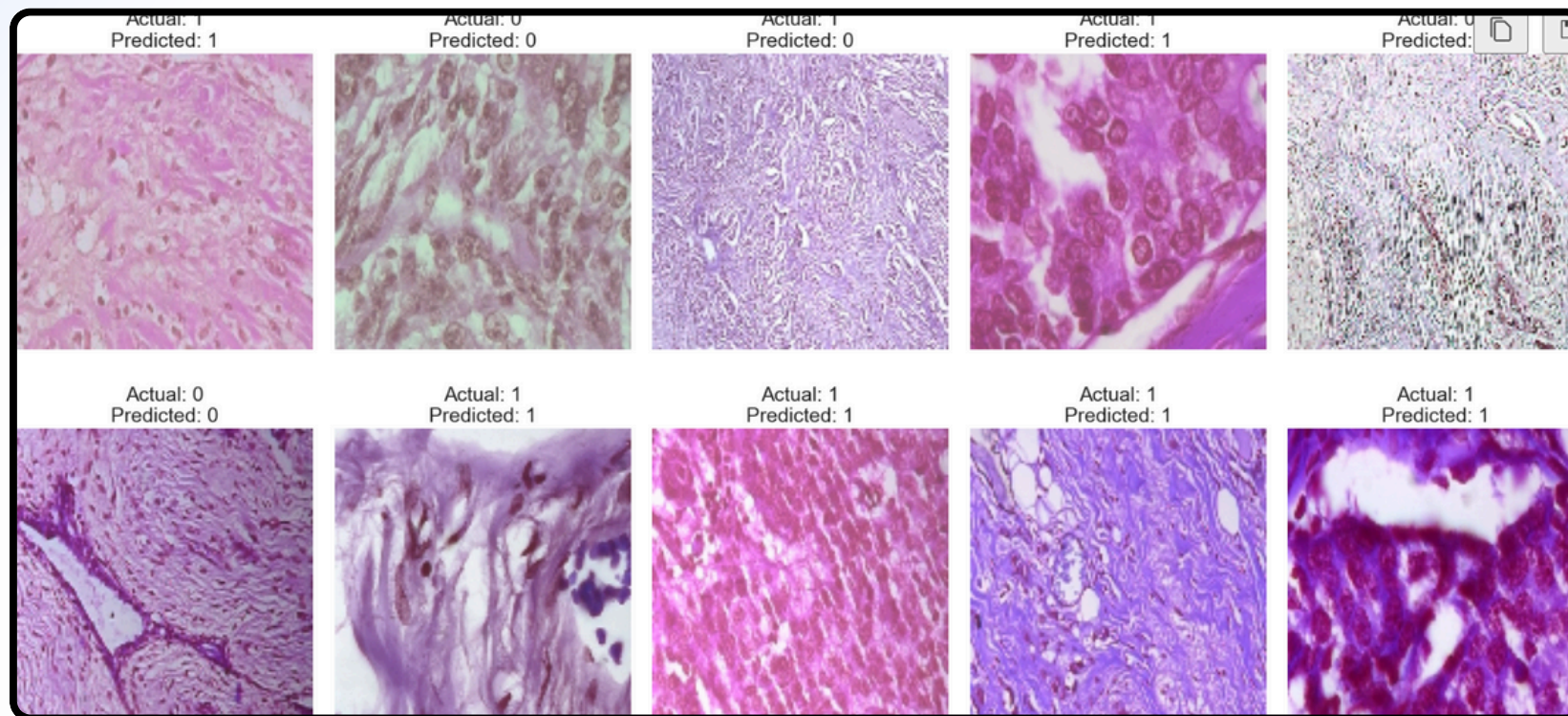
$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)} \quad p_1 \log \frac{p_1}{q_1} + p_2 \log \frac{p_2}{q_2}$$

Where:

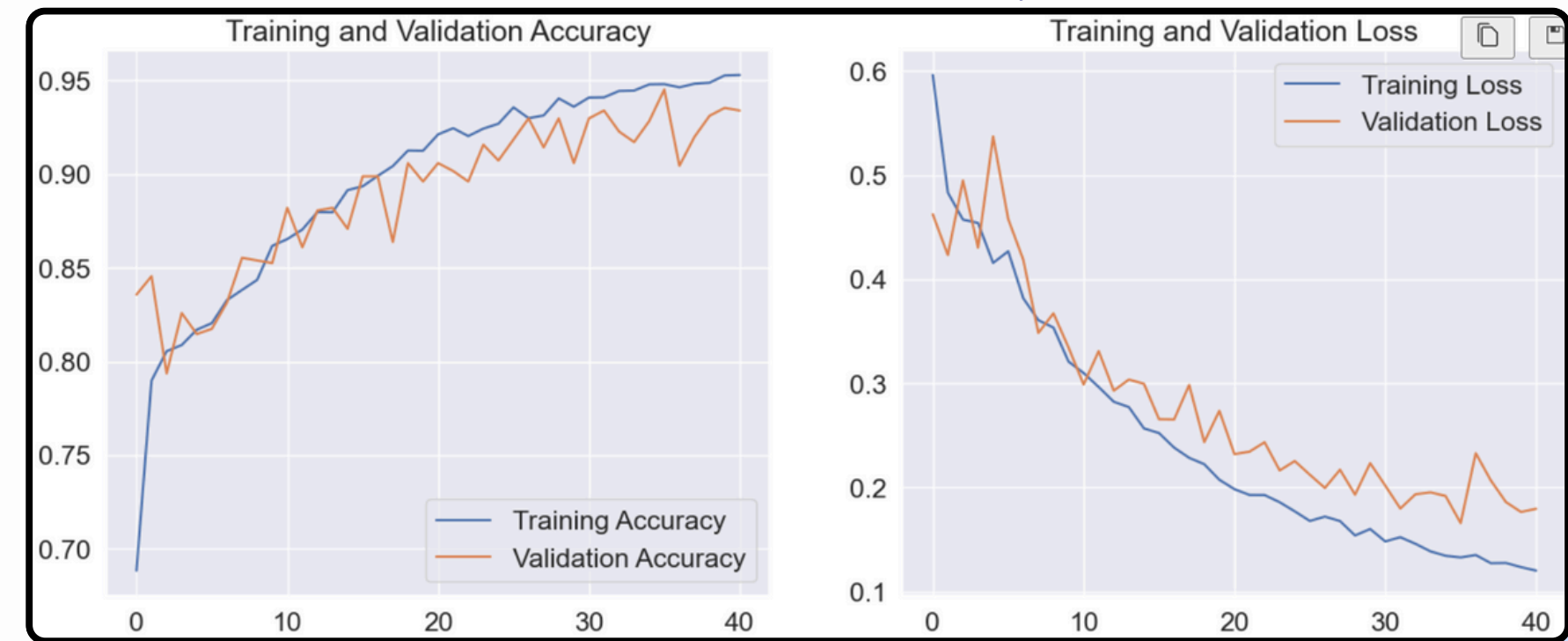
- $P(x)$ is the true (or peer) distribution
- $Q(x)$ is the predicted distribution

Results

Actual vs Prediction



Validation Accuracy/ Loss



POSTER



AI in Histopathology for Breast Cancer Classification

Om Prakash Kodande, Md. Anas Sameer Sheikh, Siddhant Nana Kate, Sagar Sanjay Mankar,
Dr. Tejashree Phatak



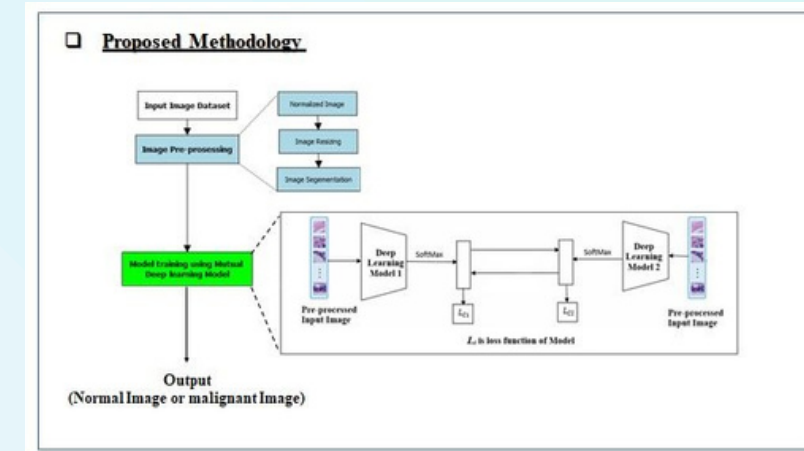
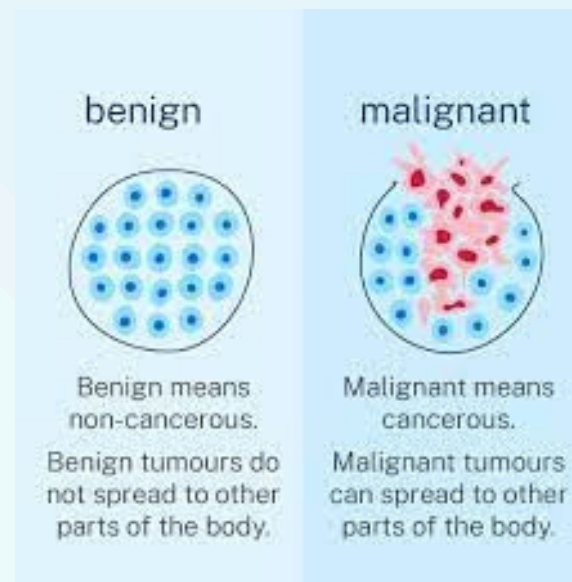
Abstract

Breast cancer is one of the leading causes of mortality among women worldwide emphasizing the need for accurate and efficient diagnostic tools. This study explores Deep Mutual Learning (DML) for histopathological breast cancer classification using Convolutional Neural Networks (CNN). leveraging KL Divergence to enhance knowledge sharing and improve classification accuracy. The dataset comprises segmented histopathology images for training and testing. Our approach demonstrates improved generalization and robustness over traditional single-model training. The results highlight DML's potential in advancing AI-driven histopathology diagnosis, reducing diagnostic errors, and aiding pathologists in early detection.

Introduction

Breast cancer remains a significant global health challenge, with early and accurate diagnosis being crucial for effective treatment and improved survival rates. Traditional histopathological analysis relies on expert pathologists, making the process time-consuming and subject to variability. Recent advancements in Artificial Intelligence (AI) and Deep Learning (DL) have shown promise in automating and enhancing diagnostic accuracy in medical imaging. In this study, we leverage Deep Mutual Learning (DML), a collaborative learning framework where two neural networks—CNN—learn from each other through Kullback-Leibler (KL) Divergence. This mutual knowledge transfer helps both models achieve improved generalization and classification performance. Our approach is applied to histopathological images, focusing on segmenting and classifying breast cancer tissue samples.

Methodology



1. Dataset Preparation

- Histopathological breast cancer images are collected and preprocessed (normalization, augmentation, and segmentation).

2. Model Selection

- Two Convolutional Neural Networks (CNNs) are used: CNN for feature extraction and classification.

3. Deep Mutual Learning (DML) :

- Both models are trained simultaneously, exchanging knowledge using Kullback-Leibler (KL) Divergence to improve learning efficiency and generalization.

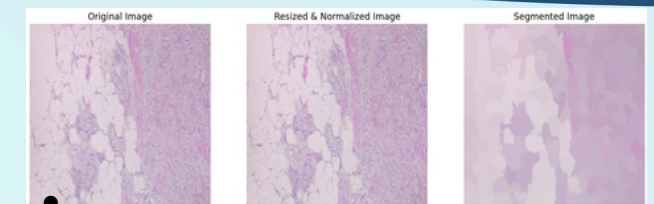
4. Training & Optimization:

- Models are trained with labeled images using cross-entropy loss and KL divergence loss.
 - a. Optimized with Adam optimizer and learning rate scheduling.

5. Evaluation & Validation:

- Performance is measured using accuracy, precision, recall, and F1-score on the test dataset. Comparison with single-model performance is conducted.
-

Results



- The proposed Deep Mutual Learning (DML) approach using CNN demonstrated improved classification accuracy compared to individual models.
- By leveraging KL Divergence, both networks effectively transferred knowledge, leading to better generalization and robustness histopathological breast cancer classification.
- Experimental results showed a significant increase in accuracy, precision, and recall, highlighting the effectiveness of DML over traditional single-model training.
- This approach enhances AI-driven diagnosis, making it a valuable tool for assisting pathologists in early breast cancer detection.

Conclusion

This study demonstrates the effectiveness of Deep Mutual Learning (DML) in histopathological breast cancer classification using CNN models. By leveraging KL Divergence, our approach enhances knowledge sharing between models, leading to improved classification accuracy and robustness. The results highlight the potential of collaborative deep learning in medical imaging, reducing diagnostic errors and supporting pathologists in early breast cancer detection.

Reference

Amandeep Kaur 1 , Chetna Kaushal 1 , Jasjeet Kaur Sandhu 1 , Robertas Damaševicius and Neetika Thakur "Histopathological Image Diagnosis for Breast Cancer Diagnosis Based on Deep Mutual Learning" Diagnostics 2024, 14, 95

Poster Competition



सह वीर्य करवावहै



Future Scope

- **Integration with Clinical Systems:** Incorporate AI models into hospital information systems for real-time analysis and decision support.
- **Multi-class Classification:** Extend the model to classify different types of breast cancer (e.g., invasive ductal carcinoma, invasive lobular carcinoma) for more detailed diagnosis.
- **Cross-Modality Imaging:** Integrate data from different imaging techniques (e.g., MRI, CT scans) to improve accuracy and provide a more holistic view.
- **Explainability in AI:** Focus on model interpretability to ensure pathologists can trust AI predictions and understand decision-making processes.
- **Wider Application:** Apply the approach to other types of cancer and medical imaging for broader diagnostic support.



THANK YOU

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